

National University of Computer and Emerging Sciences
Lahore Campus

Data Analysis for Business I Midterm 1 Exam

(M02008)

Date: April 8th, 2026.

Course Instructor(s)

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Total Time (Hrs): 1

Total Marks: 40

Total Questions: 4

98-3398

Roll No

2C

Section

Q. Waheed

Student Signature

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Attempt all the questions.

CLO # 2: Compute and interpret various measures of location and measures of variability. Able to detect outliers.

Q1: A retail company evaluates the performance of its sales team based on three criteria: Sales Volume, Customer Satisfaction, and Punctuality. Each criterion carries a different weight according to its importance. The performance details of a sales employee are given below: [5 marks]

Criteria	Score (Out of 100)	Weight (%)
Sales Volume	80	50%
Customer Satisfaction	70	30%
Punctuality	90	20%

$$x = \frac{\sum w_i x_i}{\sum w_i}$$

- Calculate the weighted mean score of the employee.
- Based on a benchmark of 75, comment on whether the employee's overall performance is satisfactory or not. Explain why weighted mean is more appropriate than simple mean in this scenario.

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.

Q2: A study was conducted to analyze whether companies provide training programs and/or performance bonuses to their employees. The following data were collected from a sample of 200 employees: [15 marks]

	Bonus: Yes	Bonus: No	Total
Training: Yes	60	40	100
Training: No	30	70	100
Total	90	110	200

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- a) Develop the joint probability table from the above data.
- b) Compute the marginal probabilities for:
 - Receiving training
 - Receiving bonus

Based on marginal probabilities, are employees more likely to receive training or bonus? Interpret your answer.

- c) If an employee received training, what is the probability that they also received a bonus?
- d) If an employee did not receive training, what is the probability that they received a bonus?
- e) Are training programs and bonuses independent? Justify your answer using probabilities.
- f) What is the probability that an employee received either training or bonus (or both)?

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.

Q3: A recent study shows that 30% of online shoppers abandon their shopping carts before completing a purchase. A sample of 15 customers is randomly selected. [10 marks]

- a) What is the probability that exactly 5 customers abandon their shopping carts?
- b) What is the probability that at least 3 customers abandon their shopping carts?
- c) If it is observed that 10 customers abandoned their carts, would you consider this result unusual? Justify your answer using probability.
- d) Compute the expected number of customers who abandon their shopping carts.

CLO # 3: Use basic counting techniques, laws of probability, specified discrete and continuous probability distributions in various business problems.

Q4: The Insurance Bureau of Canada (IBC) reports that workplace-related injury claims cost Canadian retailers over \$4.5 billion annually in compensation and lost productivity. Based on IBC estimates, retail stores with 80 employees are expected to average four workplace injury claims per year. [10 marks]

- a) What is the probability of no workplace injury claims during a one-year period?
- b) What is the probability of at least three workplace injury claims during a one-year period?
- c) What is the expected number of workplace injury claims during a three-month period?
- d) What is the probability of exactly one workplace injury claim during the next three months?
- e) What is the probability of more than two workplace injury claims during a six-month period?

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$f(0) = \frac{1^0 \cdot e^{-1}}{0!}$$

$$\frac{1}{60} \times \frac{1}{12} = \frac{1}{720}$$